

AKSHAIYA SAKTHIVEL

B.TECH – Artificial Intelligence And Data Science

OBJECTIVE

Results-driven **AI Developer and Machine Learning Engineer** with a proven record of building **production-grade trading and enterprise systems**. Expert in **machine learning, real-time data processing, and scalable machine learning model development**, delivering high-performance solutions that drive business value, automate workflows.

EXPERIENCE

AI DEVELOPER | Q DATA Tech, Abu Dhabi

AI Driven Intelligence Trading Platform (July 2025 – Present)

- **AI/ML Algorithms** : Multi-Agent analysis with **Agent-Agent MCP protocols** and **Ensemble ML models** for 85% prediction accuracy on 175+ market indicators, **sub-second inference**.
- **Backend Architecture** : **FastAPI** microservices, async processing, **SQLAlchemy**, **master-worker architecture**, REST-APIs with **JWT** auth, **MySQL** with real time audit system & backups.
- **Frontend Development**: **React-TypeScript** dashboards using TanStack Router, Live Charts and Intuitive UI, Tailwind, Shadcn/UI for **multi-account portfolio** support.
- **Infrastructure**: **MetaTrader 5 API** (500ms execution), TradingView/Polygon.io/ForexFactory real time data feeds, **Dockerized deployments**, **Telegram & NTFY** alerts.
- **Analytics & Visualization**: **100x-optimized signal engine** and automated alert reporting system, **VaR-based risk dashboards** and position sizing, backtesting frameworks.
- **Security** : **Federated-learning CNN intrusion detection**, smart risk-management for automated position closure, **Cloudflare gateway**, real-time audit monitoring.

DATA ANALYST INTERN | Heron Technologies, India

Data Pipeline Automation & Business Intelligence (May24-Mar25)

- **Frameworks**: Automated pipelines using Python and SQL, with multi-dimensional data processing by 90%.
- **Visualization**: Developed **interactive Power BI dashboards** to support data-driven decision-making.
- **Insight-Derivation**: Conducted **feature extraction and trend analysis** across large-scale datasets.

EDUCATION

NOV 2021
APR 2025

BACHELOR OF TECHNOLOGY – ARTIFICIAL
INTELLIGENCE AND DATA SCIENCE

Sri Krishna College Of Technology, India
| Coimbatore, Tamil Nadu, India



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AI Danah, Abu Dhabi



Apr 5, 2003



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SKILLS

- ✎ **Cloud & Devops** : AWS, Azure, Kubernetes, Docker, CI/CD Pipelines,
- ✎ **LLMOps & Model Lifecycle**: MLflow, Weights & Biases, DVC, Feast, Airflow/Prefect, Kubeflow Pipelines, Kubernetes, Seldon Core/BentoML, Ray, Prometheus, Grafana, Evidently AI
- ✎ **AI & Machine Learning**: TensorFlow, CNNs, Federated Learning, LGBM, LSTM, Feed Forward Neural Networks, Langchain, Langgraph, Transformers
- ✎ **Data Analytics** : SQL Querying, Python (Pandas, NumPy), Power BI, Feature Extraction, Trend Analysis, Report Generation
- ✎ **API Development & Security**: REST API, API Gateway, JWT Authentication, Spring Boot, Microservices, SQLAlchemy, FastAPI
- ✎ **Testing & Automation Tools**: Selenium IDE, Postman, TestNG Framework
- ✎ **Programming Languages** : Python, Java, C++, HTML/CSS, JavaScript, React JS, Tailwind, MQL, RDBMS, PostgreSQL
- ✎ **Trading & Finance**: MT5, TradingView APIs, Risk management, Algorithmic Trading, Stop Protection, Automatic Stop-Loss and TakeProfit Threshold Detection and News-

PROJECTS

AI Trading Intelligence Platform

Real-Time Market Prediction - Machine Learning Model

- **Multi-agent AI system** with technical, fundamental, and sentiment analysis engines to feed real time data - AI Model
- **Ensemble ML models (LSTM, FFNN, LightGBM)** achieving 85%+ prediction accuracy and prediction of trend pattern
- **Feature Engineering** from multiple features of technical, fundamental, sentimental indicators and correlation matrix with real-time market data processing using **Python**
- **Analytics Dashboard** – Interactive UI to visualize the performances by account-wise, symbol-wise, session-wise historical and live data comparison metrics

Signalzyzer - Trading Signal Analysis Platform

Automated Multi-Signal Mapping & Execution Software

- **High-Performance optimization** using multiple matching algorithms to map unique execution of unique signals from webhook to multiple MT5 accounts at a time
- **Multi-source integration** processing trading view signals in any format (Raw Json, Gmail, CSV) and webhook executions
- **CLI automation** tools for signal verification and analysis workflows with
- **JWT authentication** and async background task processing for parallel processing of dynamic data

Real-Time Trading Audit System

24/7 Risk Management & Automated Monitoring System

- **Comprehensive audit rules** across multiple risk categories for live trading accounts
- **Multi-channel alert system** with Telegram Bot and Ntfy push notifications and real-time dashboard alerts
- **30-second monitoring cycles** with anti-spam protection and graceful error handling and
- **Automated profit/loss threshold** monitoring preventing financial losses and auto emergency activate protocols

Federated Deep Learning Approach For Intrusion Monitoring

Privacy-Preserving Cybersecurity with CNNs

- **Federated learning architecture** for distributed threat detection across network infrastructures
- **CNN and RNN models** for advanced threat detection including DDoS attacks
- **Python-based modular design** with seamless deployment capabilities

CERTIFICATIONS & COURSES

-  IBM Python For Data Science
-  Fundamentals Of Artificial Intelligence
-  MATHLAB Machine Learning
-  CISCO Cybersecurity
-  Data Analysis With
-  Robotic Process
-  Scientific Computing With Python
-  2D-3D CAD Modeling of Building Design
-  Front End Development
-  Network And Communications
-  HackerRank Java Certification
-  HackerRank Problem Solving
-  IAMNeo Best Project Certification

PERSONAL DETAILS

- **Nationality** : Indian
- **Gender** : Female
- **Passport Number** : U8814046
- (Valid Upto : 23.02.2031)
- **Emirates ID** : 784-2003-4277562-2
- (Valid Upto : 17.04.2027)

LANGUAGES

- English • Arabic • Hindi • Tamil